

Implicit Differentiation and Inverse Functions



Implicit differentiation basics

- 1 Find $\frac{dy}{dx}$ for $x^2 + y^2 = 25$.
 - 2 Find $\frac{dy}{dx}$ for $x^3 + y^3 = 6xy$.
 - 3 Find $\frac{dy}{dx}$ for $\sin(xy) = x$.
-

Tangent lines to implicit curves

- 4 Find the equation of the tangent line to $x^2 + y^2 = 25$ at the point $(3, 4)$.
 - 5 Find $\frac{dy}{dx}$ at the point $(1, 2)$ on the curve $x^2y + y^3 = 10$.
-

Second derivatives via implicit differentiation

- 6 Find $\frac{d^2y}{dx^2}$ for $x^2 + y^2 = 1$, expressing your answer in terms of x and y .
-

Derivatives of inverse functions

- 7 If $f(x) = x^3 + 2x + 1$, find $(f^{-1})'(4)$, using $(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$ and noting $f(1) = 4$.
 - 8 State the derivatives of the inverse trigonometric functions $\sin^{-1}x$, $\cos^{-1}x$, and $\tan^{-1}x$.
 - 9 Differentiate $f(x) = \tan^{-1}(3x)$.
-

Implicit differentiation with logarithmic and exponential terms

- 10 Find $\frac{dy}{dx}$ for $e^y = xy$.
 - 11 Find $\frac{dy}{dx}$ for $\ln(x + y) = x^2$.
-

Related applications of implicit differentiation

- 12 Find all points on the curve $x^2 + xy + y^2 = 7$ where the tangent line is horizontal.
-

Finding vertical tangents implicitly

- 13 Find the points on the circle $x^2 + y^2 = 9$ where the tangent line is vertical (i.e. where $\frac{dx}{dy} = 0$, or equivalently where $\frac{dy}{dx}$ is undefined).
-

Inverse function derivatives from a graph or table

- 14 A function f is one-to-one and differentiable, with $f(3) = 10$ and $f'(3) = 4$. Find $(f^{-1})'(10)$.
- 15 If $g(x) = x^5 + 2x - 3$, find $(g^{-1})'(-3)$, first identifying the value a such that $g(a) = -3$.
-

Implicit differentiation in an applied context

- 16 A conical tank has height and radius related by $h = 2r$ at all times as it fills, so its volume is $V = \frac{1}{3}\pi r^2(2r) = \frac{2}{3}\pi r^3$. Using implicit differentiation with respect to time, find $\frac{dV}{dt}$ in terms of r and $\frac{dr}{dt}$.
-

A capstone implicit differentiation problem

- 17 Find $\frac{dy}{dx}$ for the curve $x^2y^2 + xy = 6$ at the point $(1, 2)$.
- 18 Find $\frac{dy}{dx}$ for $\cos(x - y) = y\sin x$.
- 19 A satellite dish's cross-section satisfies $4x^2 + y^2 = 16$ (an ellipse). Find $\frac{dy}{dx}$ at the point $(1, 2\sqrt{3})$ on the curve.
- 20 For the curve $y^2 = x^3$ (a cuspidal cubic), complete each step:
- a Find $\frac{dy}{dx}$ implicitly.
 - b Evaluate $\frac{dy}{dx}$ at the point $(4, 8)$.
 - c Find the equation of the tangent line at that point.
- 21 If $f(x) = 2x^3 + x - 5$ is one-to-one and differentiable, find $(f^{-1})'(-2)$, first identifying the input a such that $f(a) = -2$.

22 For the hyperbola $xy = 6$, complete each step:

- a** Find $\frac{dy}{dx}$ using implicit differentiation.
- b** Confirm your answer matches what you'd get by first solving for $y = \frac{6}{x}$ and differentiating directly.
- c** Evaluate $\frac{dy}{dx}$ at the point $(2, 3)$.